

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_isolado.parquet

model_type: LSTM
group_sec: 200
timesteps_orig: 60
batch_size: 512
min_units: 16
max_units: 512
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 100
hyperband_iterations: 3
group_msec: 200
samples_per_sec: 5
forecast_horizon_sec: 4
forecast_steps: 20
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-08-01 16:16:46

Training Start: 2025-08-01 16-10-36

Training End: 2025-08-01 16-15-56

Training Duration: 5.34 minutes

Predictions Start: 2025-08-01 16-15-56

Predictions End: 2025-08-01 16-16-11

Predictions Duration: 0.24 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: elu

Optimizer: adam

Dropout Rate: 0.20

MAE: 2.5569

MSE: 9.3920

RMSE: 3.0646

R2 Score: -0.3457

Multi-Step Forecast:

Forecast Steps: 20

MAE: 0.0620

MSE: 0.0061

RMSE: 0.0780

R2 Score: 0.6021

Dataset size before down-sample: 31,097 rows
Dataset size after down-sample: 1,870 rows

First 5 timestamps BEFORE:

1744464597710
1744464597720
1744464597744
1744464597757
1744464597783

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

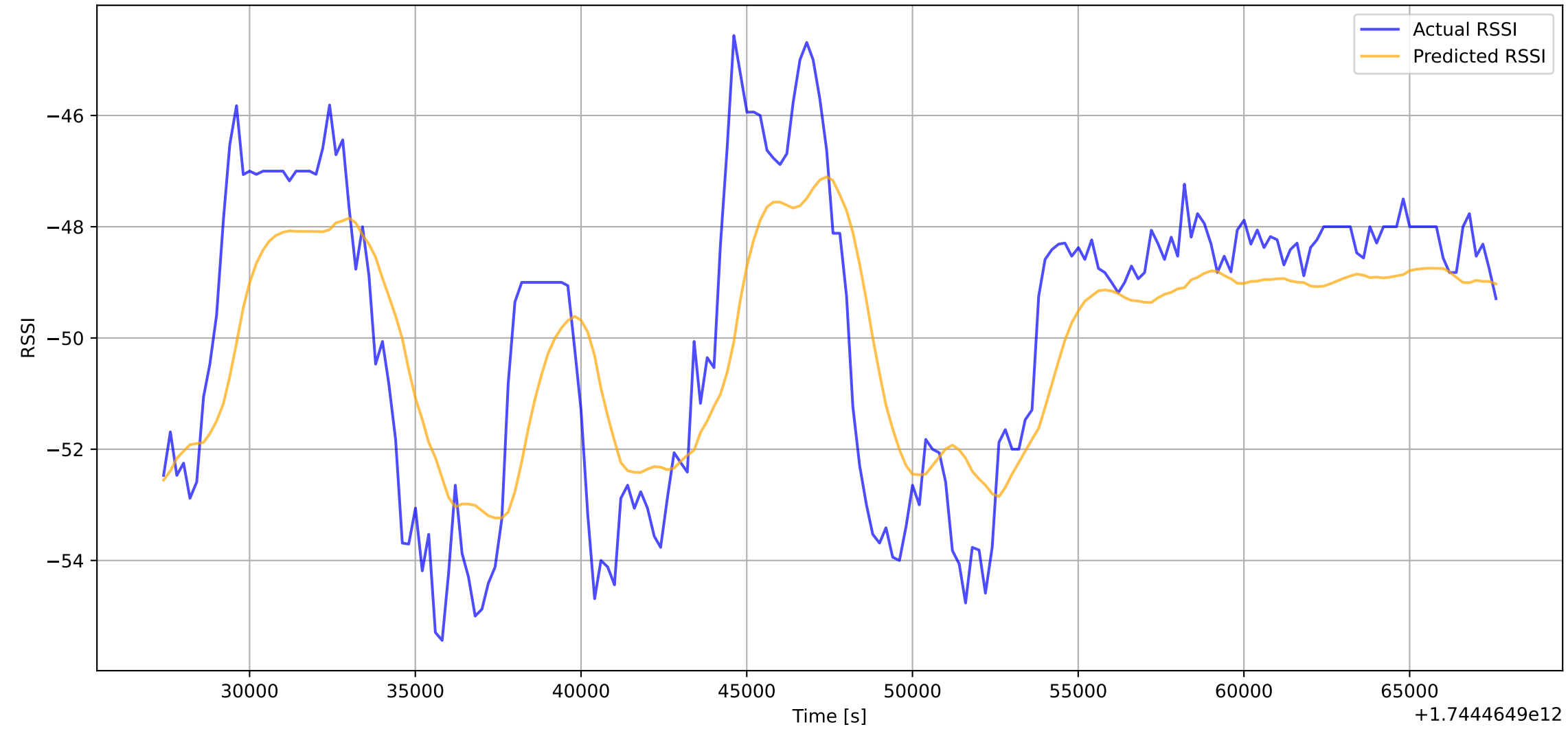
1744464597710
1744464597812
1744464598008
1744464598210
1744464598404

...

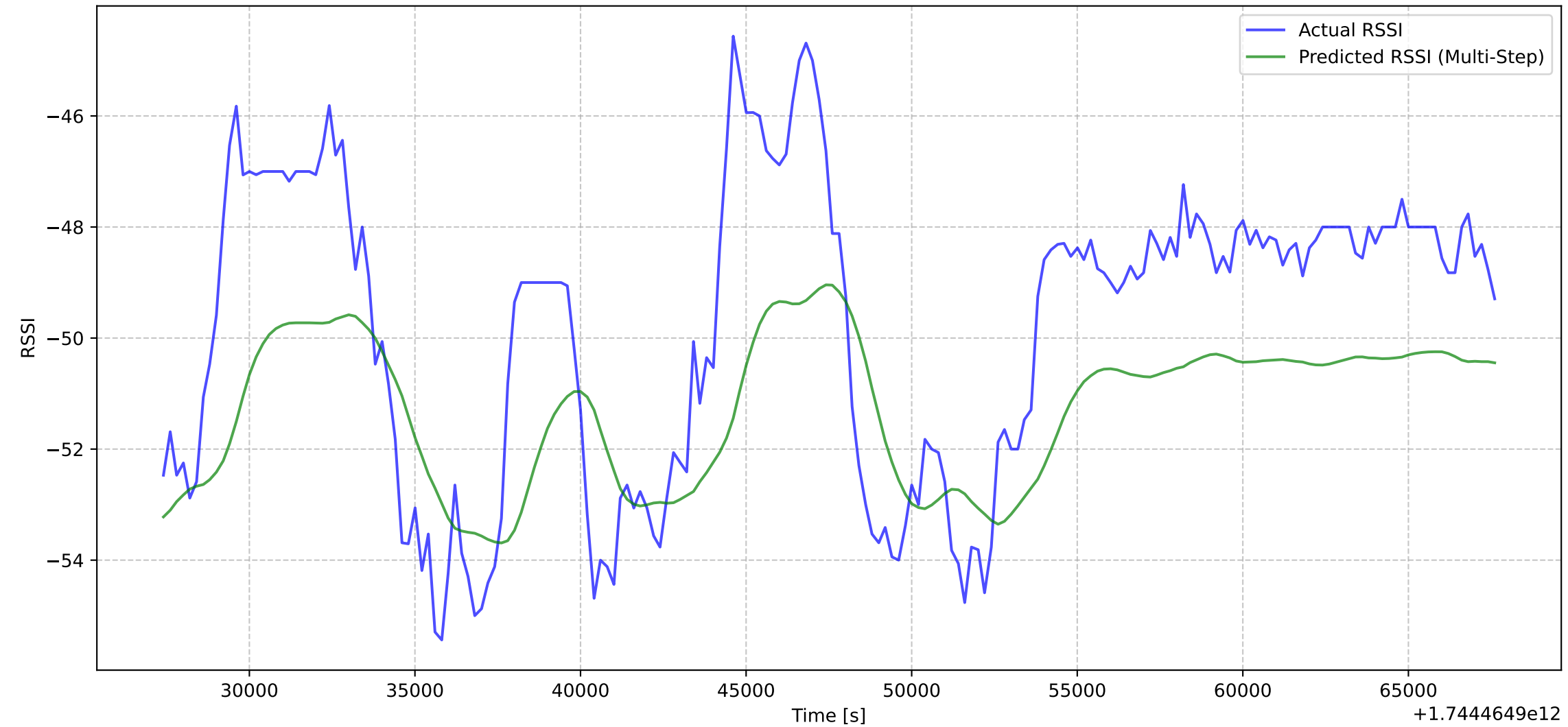
Last 5 timestamps AFTER:

1744464970609
1744464970802
1744464971004
1744464971209
1744464971402

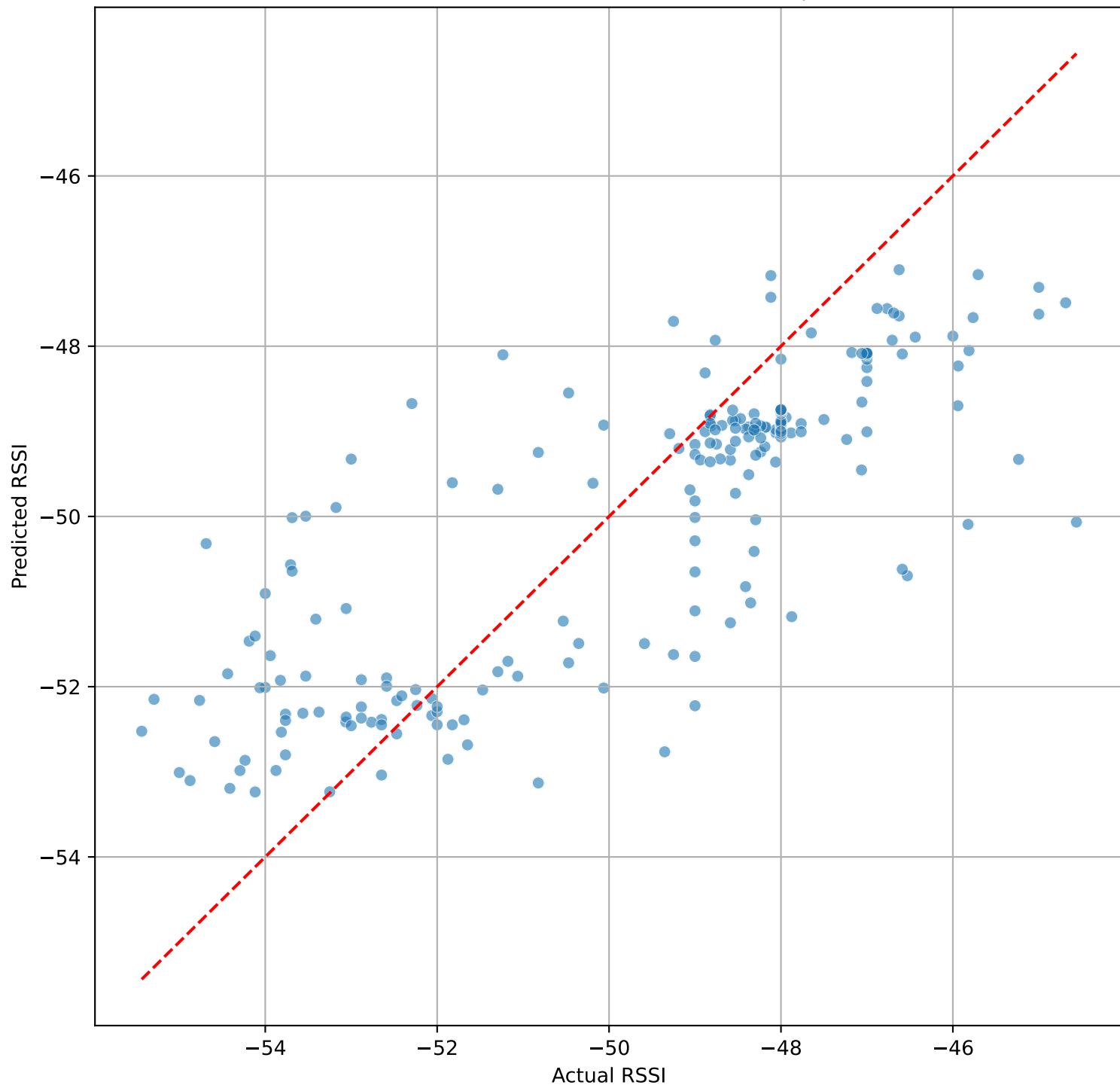
Actual vs Predicted RSSI (First Step)



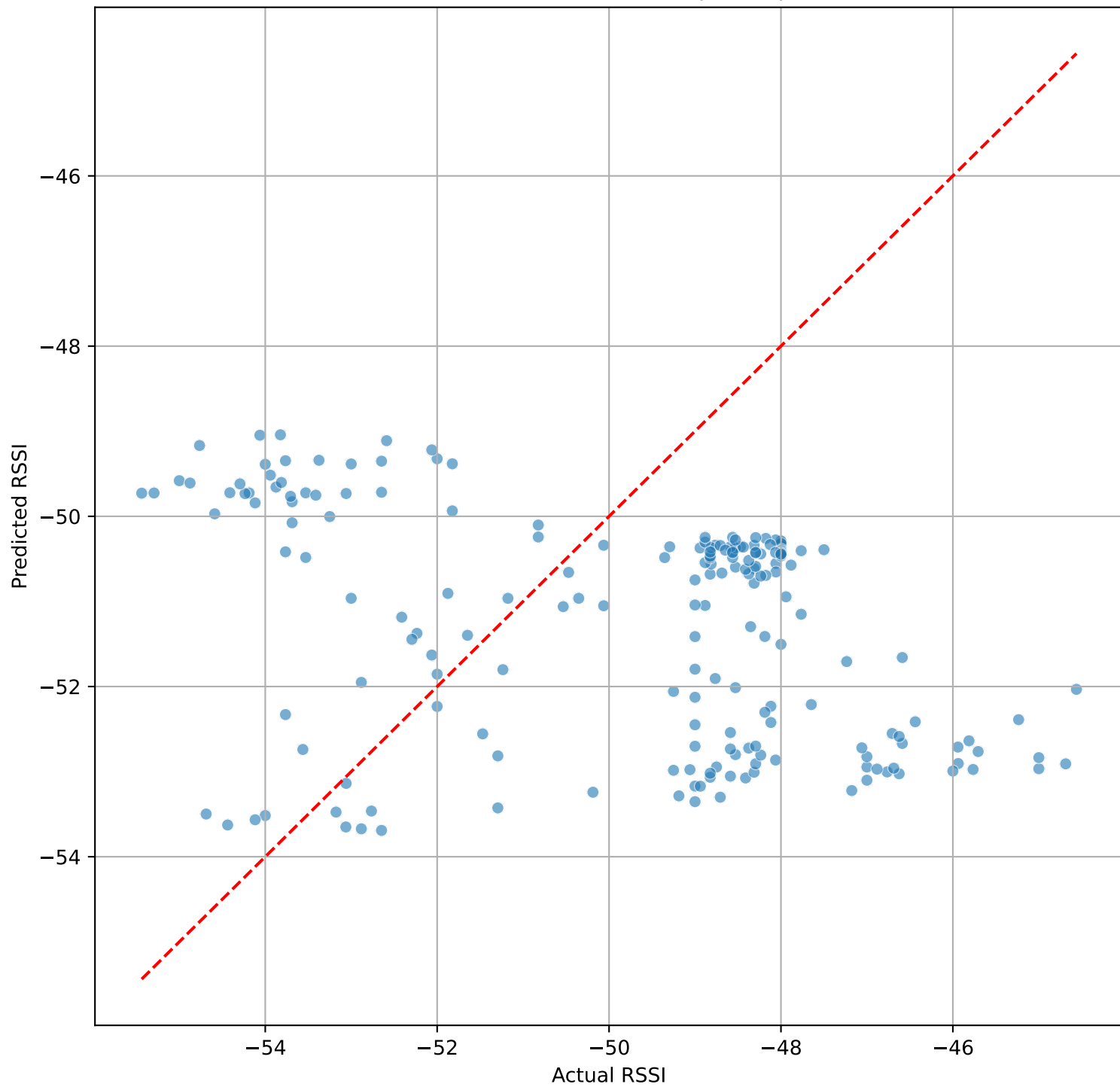
Actual vs Predicted RSSI (Multi-Step (step 20 $\approx$ 4.0 s))



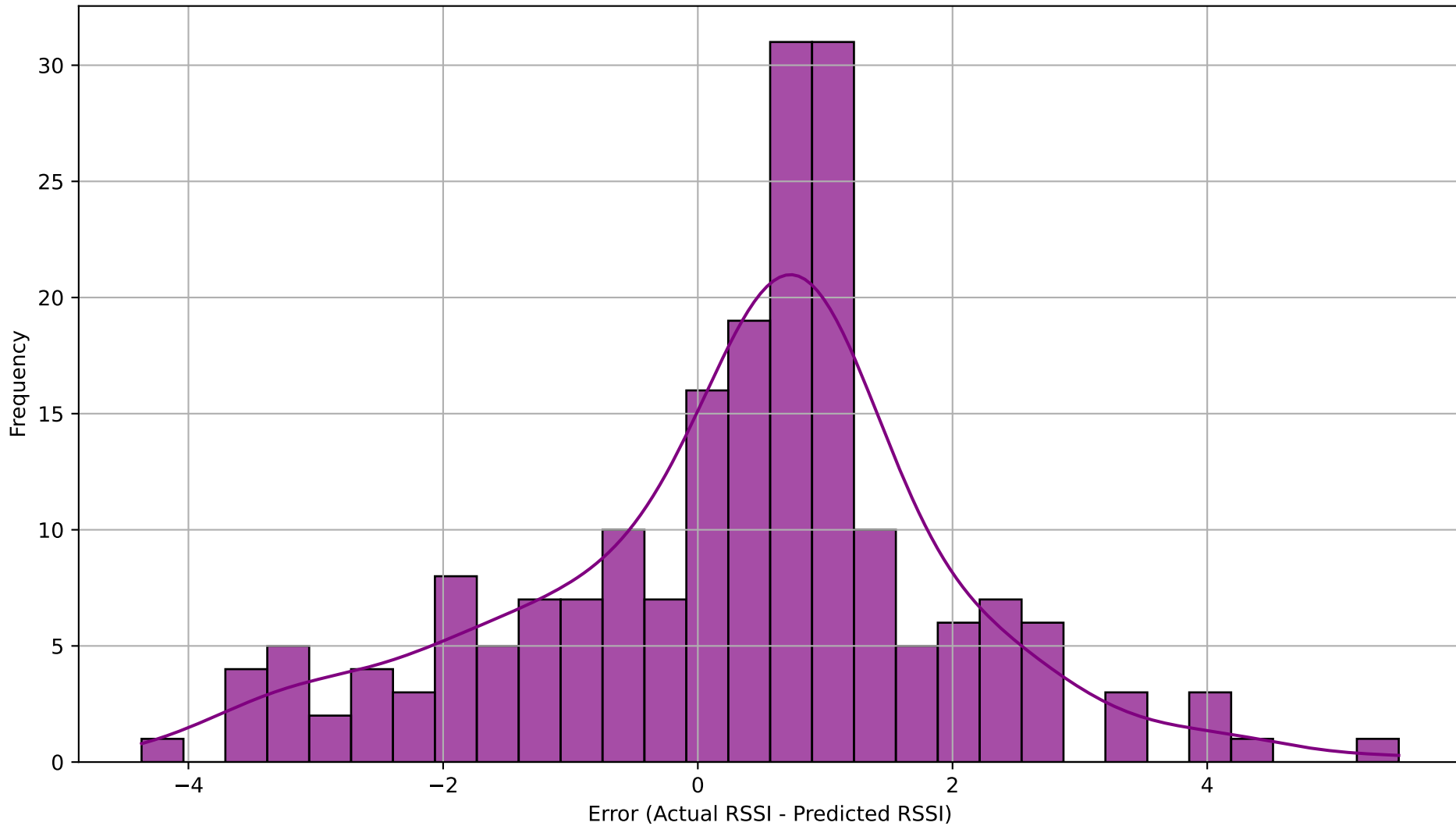
Actual vs Predicted RSSI (First Step)



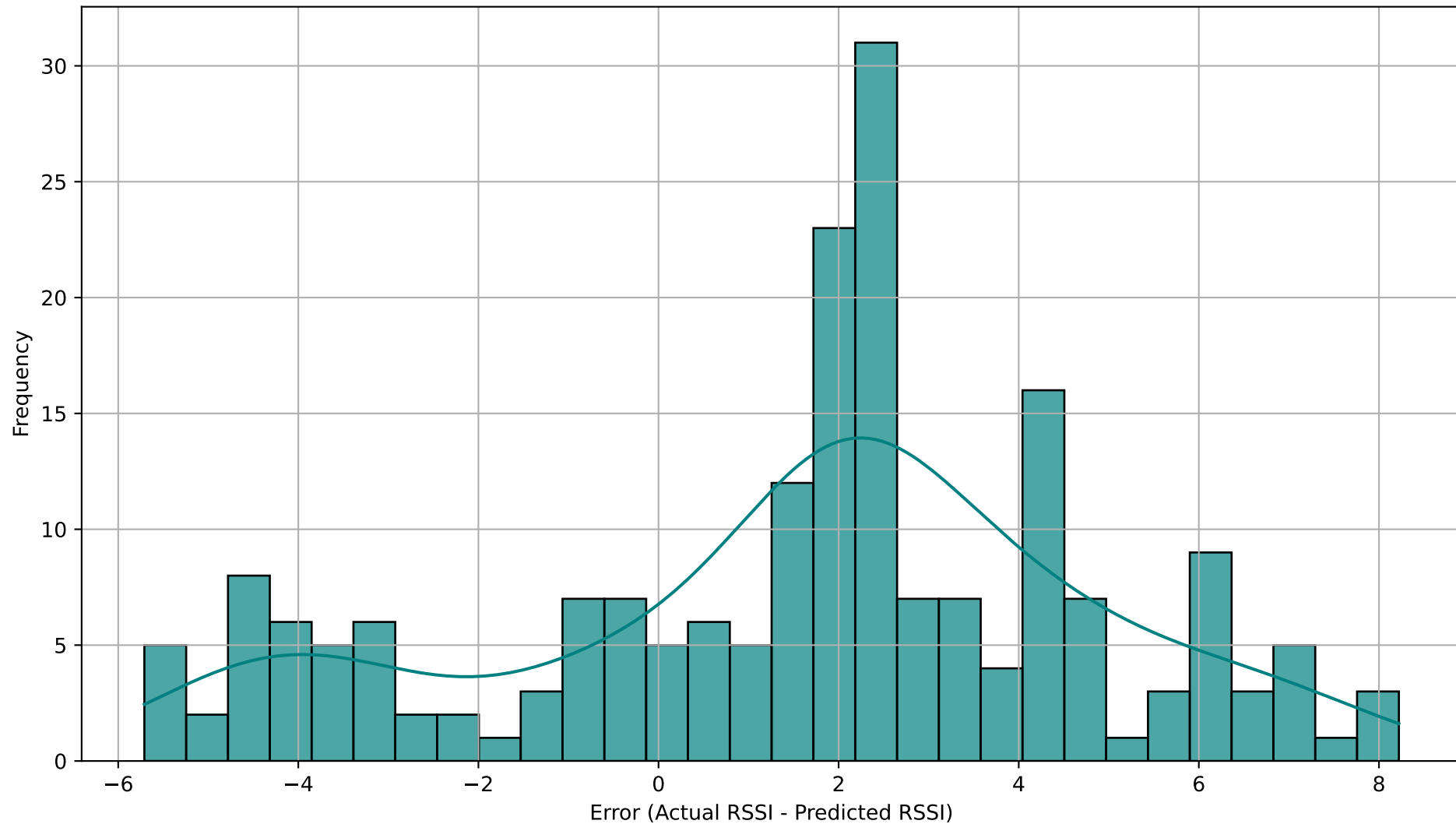
Actual vs Predicted RSSI (Last Step - step 20 $\approx$ 4.0 s)



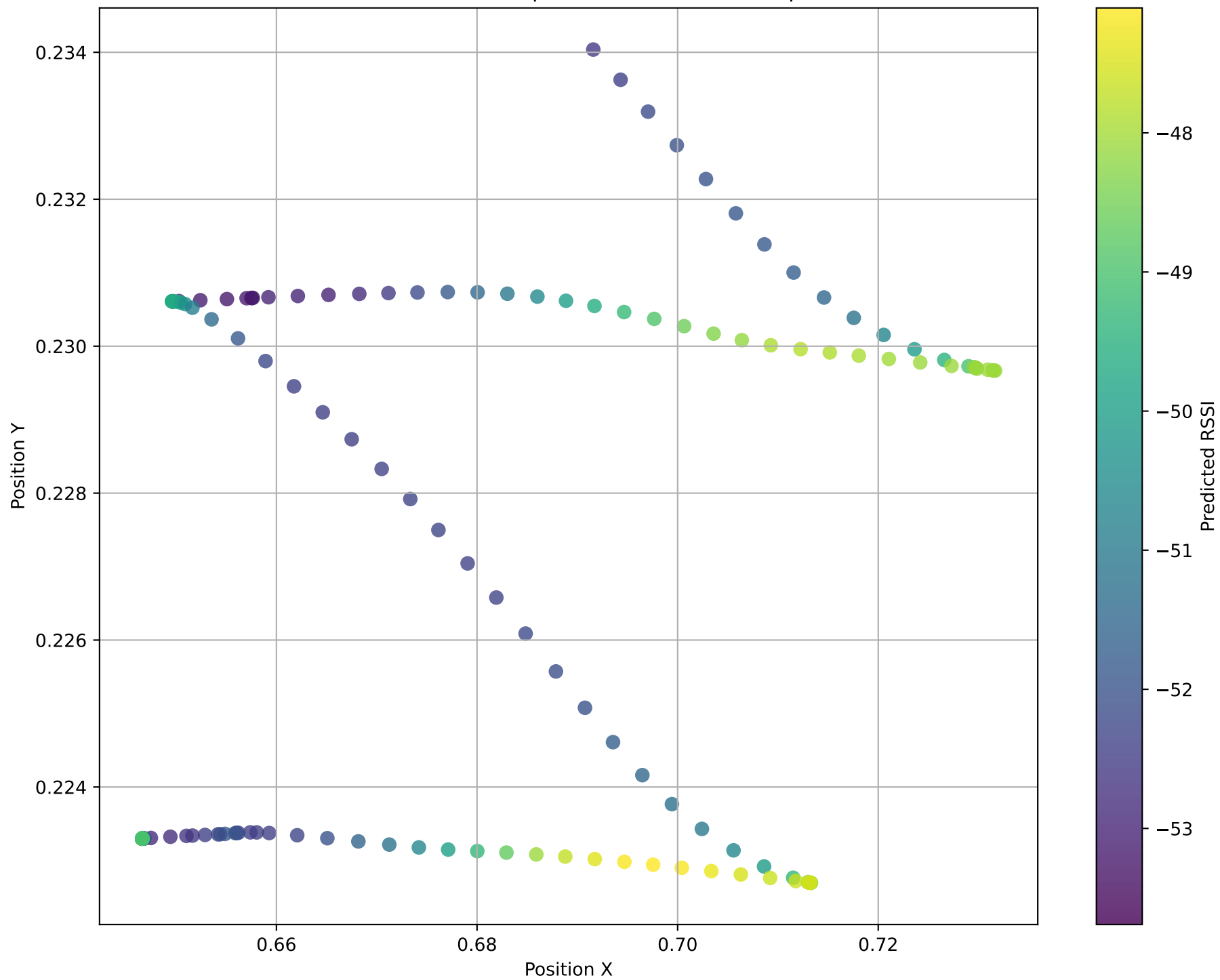
Distribution of Prediction Errors (First Step)



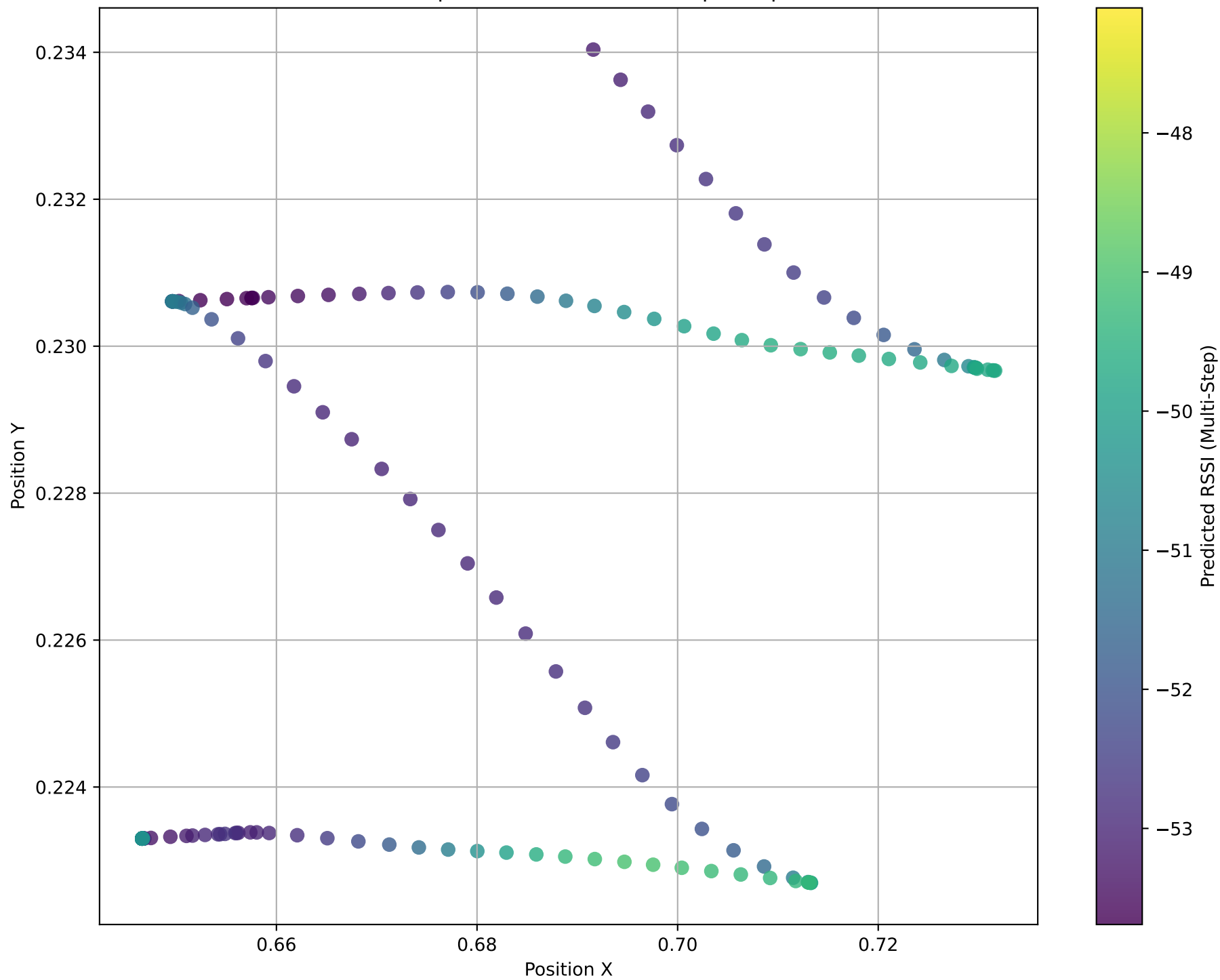
Distribution of Prediction Errors (Last Step - step 20 $\approx$ 4.0 s)



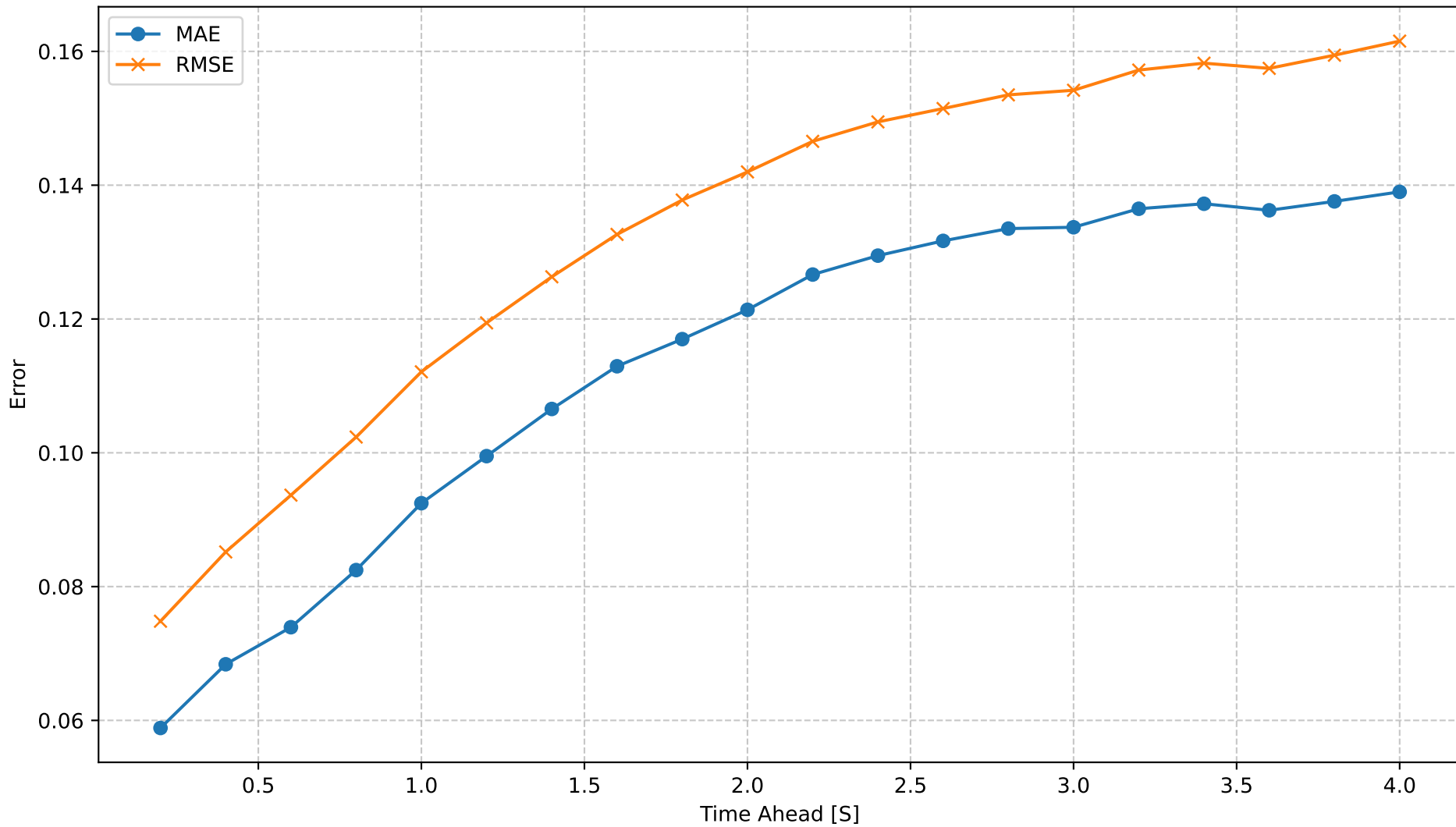
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_isolado.parquet
Modelo: LSTM
Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.06 / 0.07
- Mediana : 0.12 / 0.14
- 20º passo : 0.14 / 0.16

Crescimento do erro: 136.1% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 20 (0.14), 19 (0.14), 17 (0.14)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.